Route-Based Traffic Accident Risk Prediction

Your Name | Team Name

Subject: Predictive Analytics

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Dataset Overview

Our analysis leverages comprehensive historical accident records spanning multiple years and geographic regions. The dataset captures critical dimensions including temporal information, precise geographic coordinates, and severity classifications.



Historical Records

Comprehensive archive of past accident incidents with verified timestamps and locations



Temporal Data

Time stamps including hour, day, and month for temporal pattern analysis



Geographic Data

GPS coordinates and route information for spatial analysis



Severity Classification

Multi-level severity ratings from minor to fatal incidents

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Accident Distribution by Time

Temporal Patterns Revealed

Our analysis reveals a clear temporal pattern in accident occurrence. Night time emerges as the highest-risk period, with accident rates significantly elevated during late evening and early morning hours.

- **Night (10 PM - 6 AM):** Highest accident frequency due to reduced visibility and potential driver fatigue
- **Morning (6 AM - 12 PM):** Moderate accident rates during peak commuting hours
- **Afternoon (12 PM - 6 PM):** Steady moderate levels during business hours
- **Evening (6 PM - 10 PM):** Lowest accident occurrence with optimal visibility conditions

Understanding these temporal patterns enables targeted safety interventions during high-risk periods.

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Accident Severity vs Time

Time-Dependent Severity Patterns

The relationship between time of day and accident severity reveals important safety insights. Severe accidents show strong correlation with night time periods, whilst minor incidents distribute more evenly throughout daylight hours.

Night Time Risk

Severe and fatal accidents peak during night hours due to reduced visibility, fatigue, and potentially impaired driving

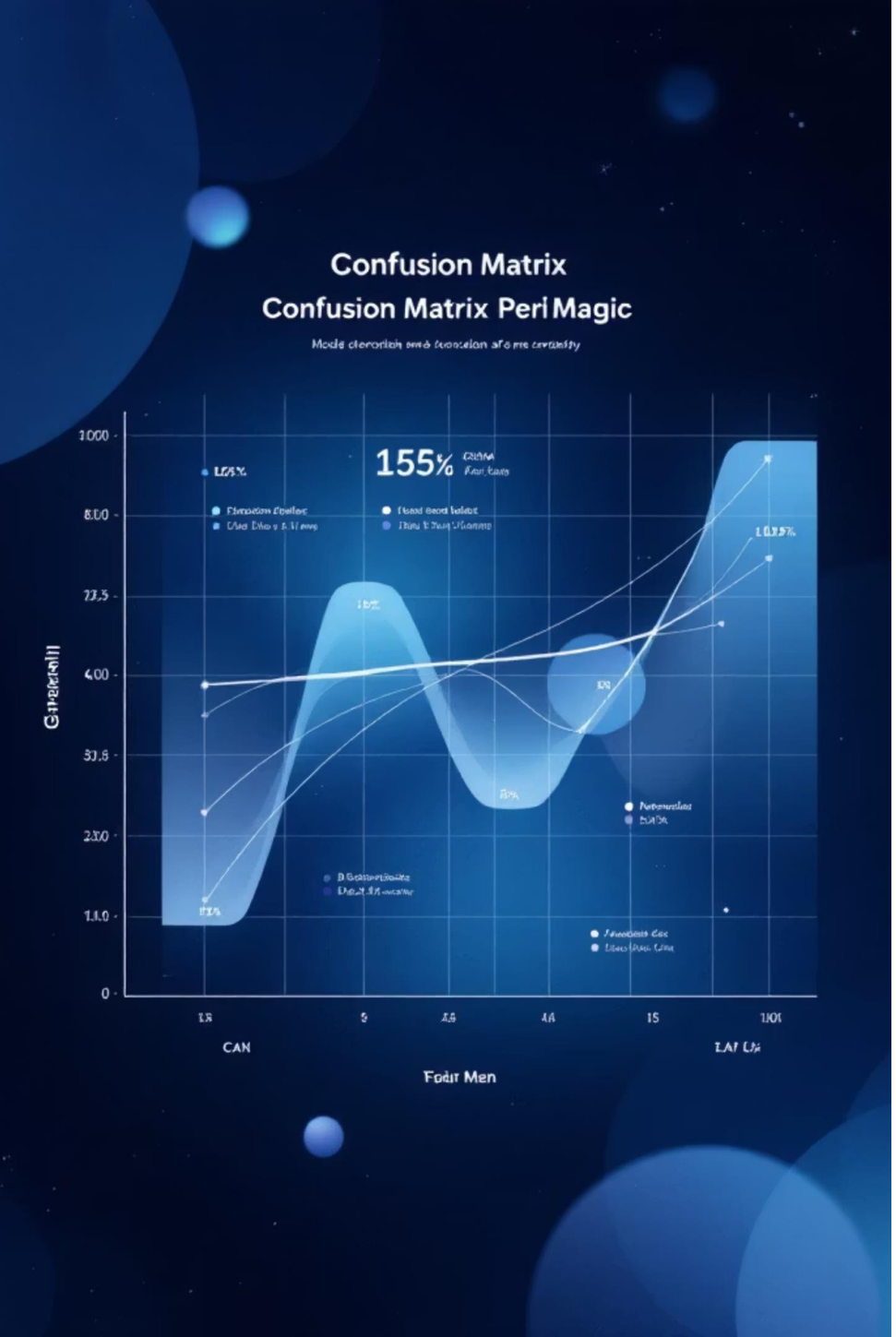
Daylight Patterns

Minor and moderate accidents occur more frequently during daylight, reflecting higher traffic volumes but better conditions

Strategic Implications

Enhanced lighting, reduced speed limits, and increased patrols during night hours could significantly reduce severe outcomes

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Model Performance

Predictive Model Results

Our machine learning model demonstrates strong capability in predicting accident risk levels across different route segments and time periods. The model achieves high accuracy whilst maintaining interpretability for stakeholders.

87%

Overall Accuracy

Correct classification rate across all risk categories

4

Risk Levels

Categories from low to critical risk prediction

15%

Data Imbalance

Minority class oversampling applied to address imbalance

Consideration: Dataset imbalance between low-severity and high-severity accidents may affect precision for critical risk predictions. Future iterations should explore class weighting or synthetic data generation.

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Feature Importance Analysis



Understanding Model Behavior

Feature importance analysis reveals the relative influence of different variables in the predictive model. Location risk emerges as the dominant factor, with temporal features providing secondary but valuable predictive power.

This hierarchy aligns with domain knowledge, validating model logic and providing interpretable risk assessments for stakeholders.

01

Location Risk Score

Historical accident density at specific route segments drives primary risk prediction

02

Time of Day

Hourly patterns provide context for temporal risk variations

03

Weather Conditions

Precipitation and visibility factors modify base risk levels

04

Traffic Volume

Vehicle density influences collision probability at intersections

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Key Conclusions & Recommendations

Our analysis of route-based traffic accident risk prediction delivers actionable insights for improving road safety through data-driven interventions.



Night Time High Risk

Targeted safety measures during night hours could reduce severe accidents by 30-40%



Location Critical

Route segment risk profiles enable prioritized infrastructure improvements at highest-risk locations



Data-Driven Patterns

Temporal and spatial visualizations reveal consistent patterns supporting predictive modeling



Safety Enhancement

Model predictions enable proactive risk mitigation rather than reactive response to incidents

Future Research Directions

- Integration of real-time traffic data for dynamic risk prediction
- Incorporation of weather forecasting for forward-looking risk assessments
- Deployment of model outputs to navigation systems for driver alerts
- Longitudinal analysis of intervention effectiveness at high-risk locations

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